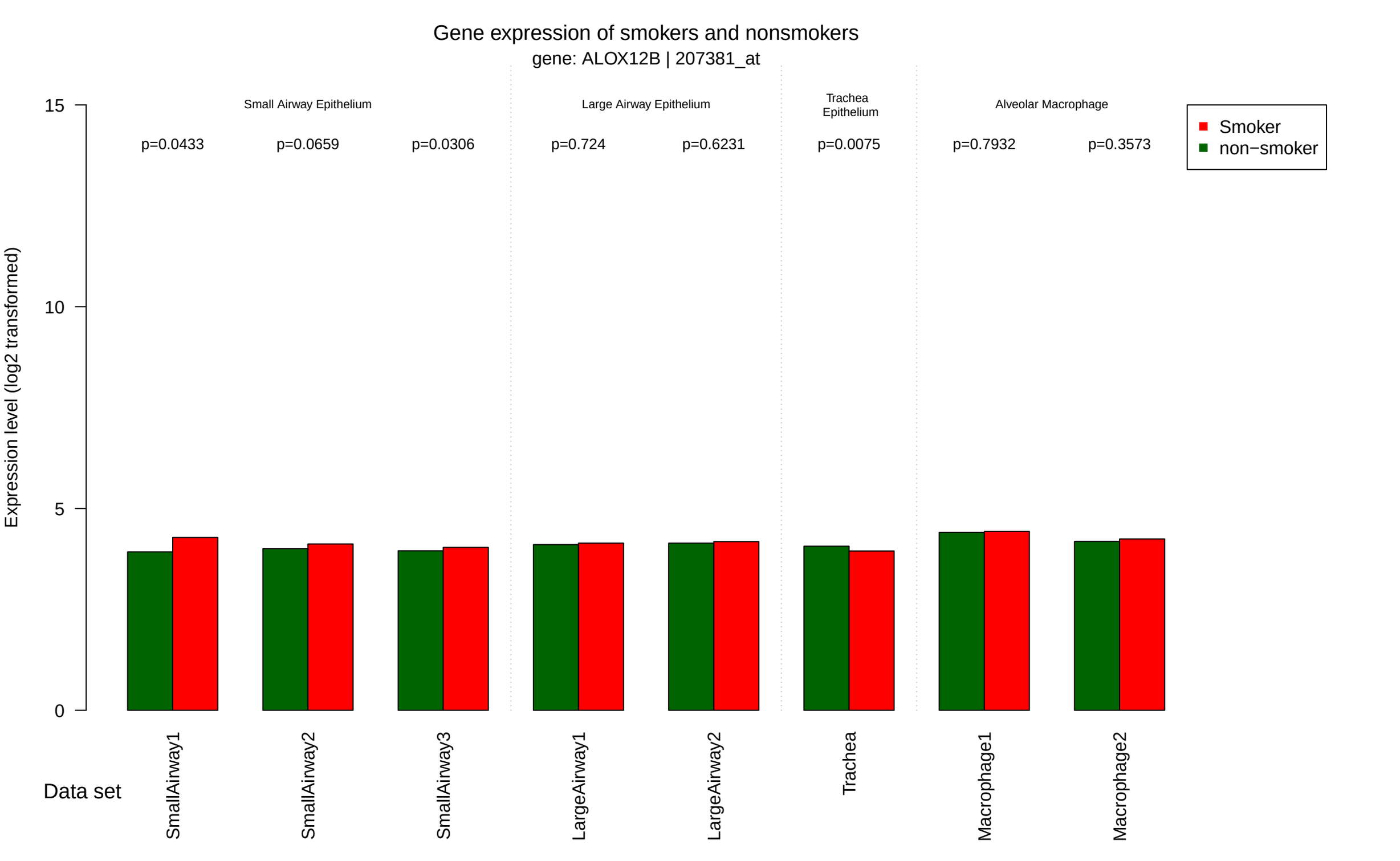
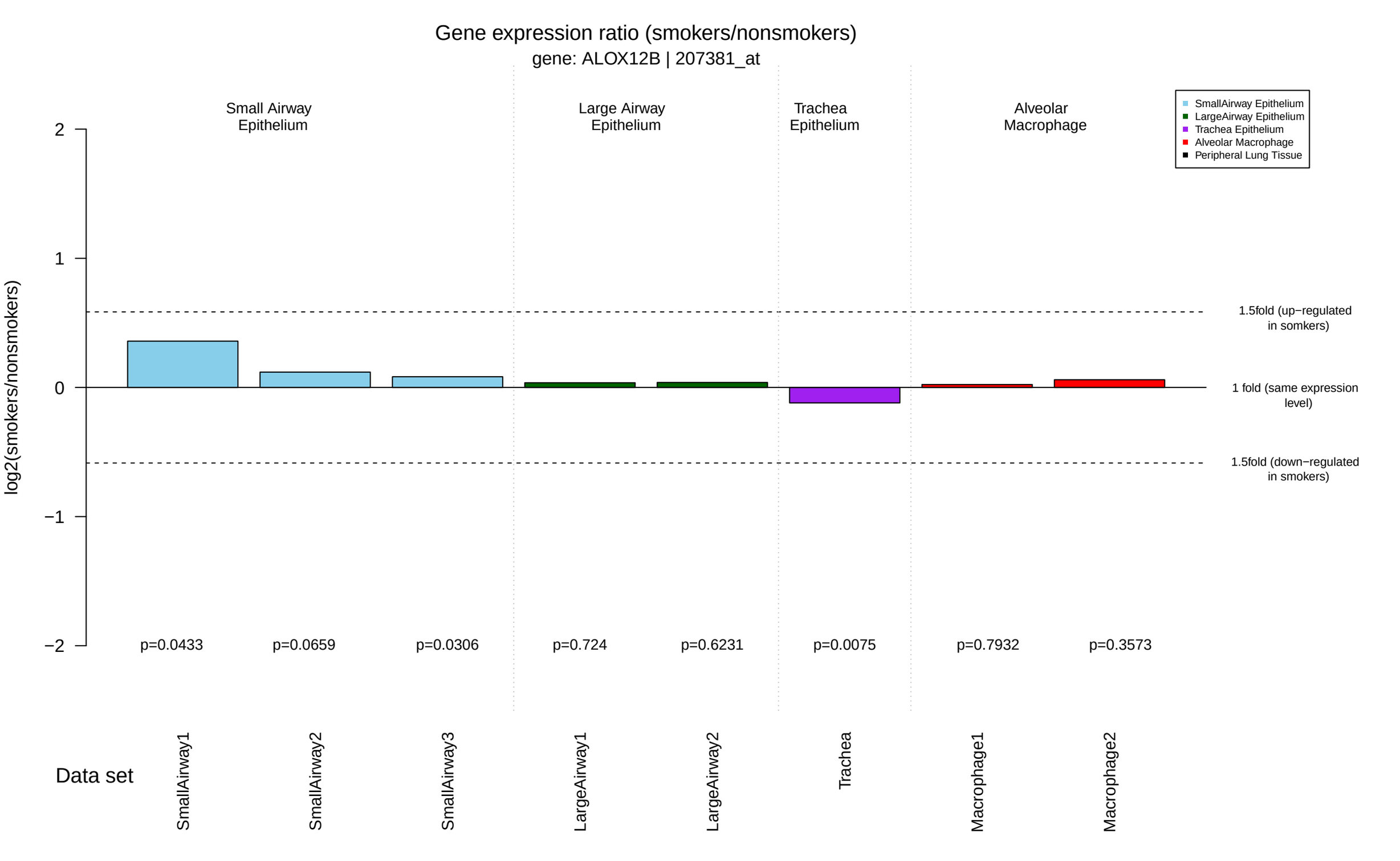
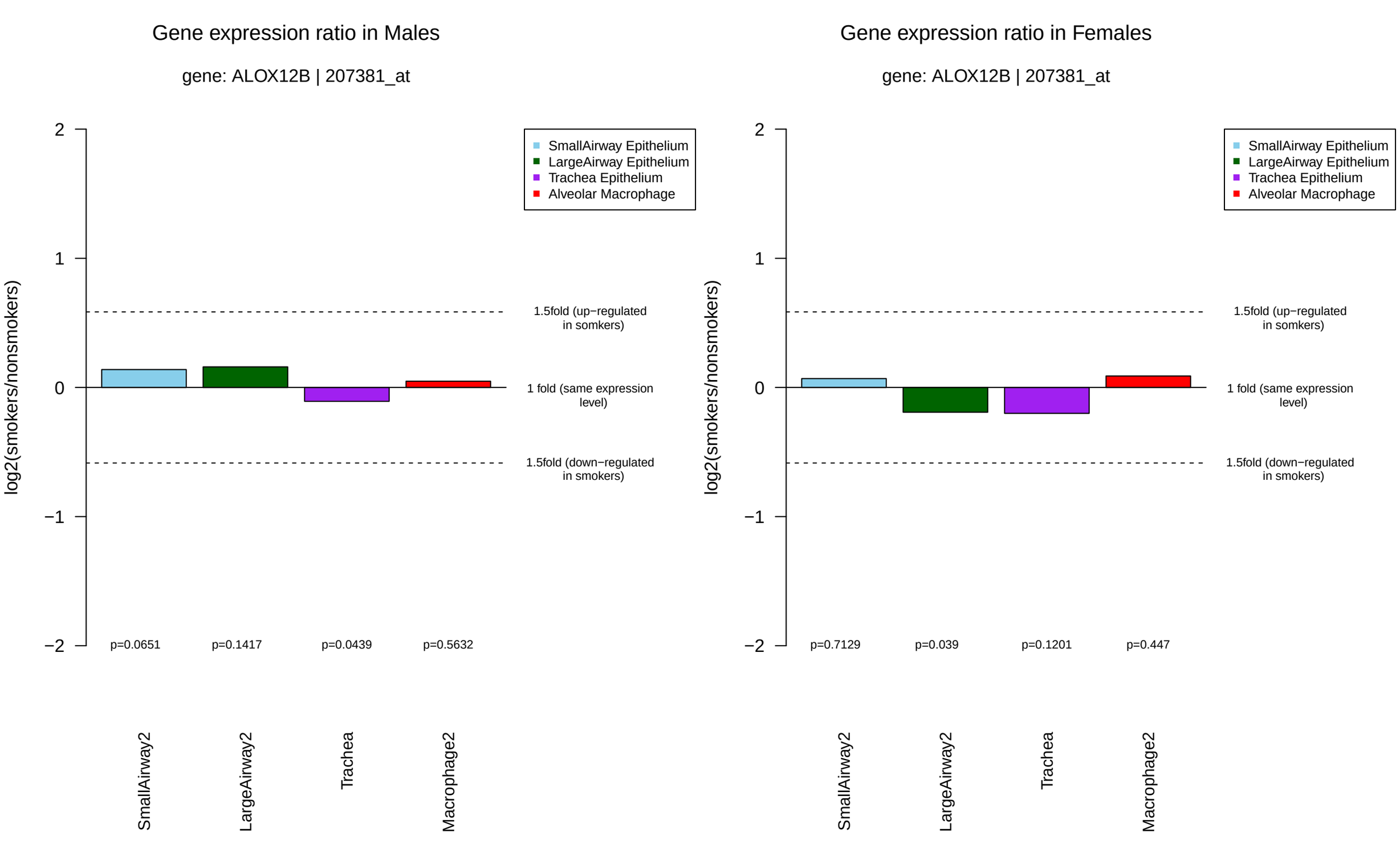
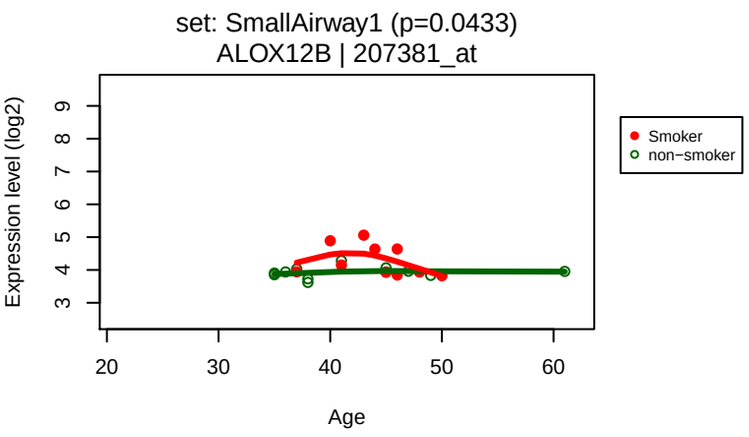
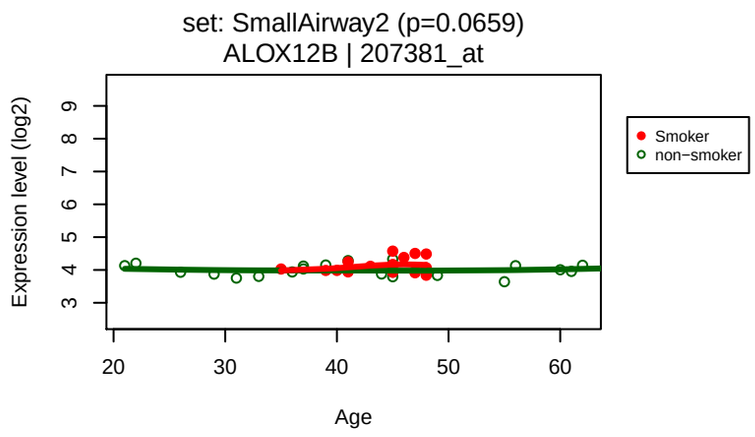
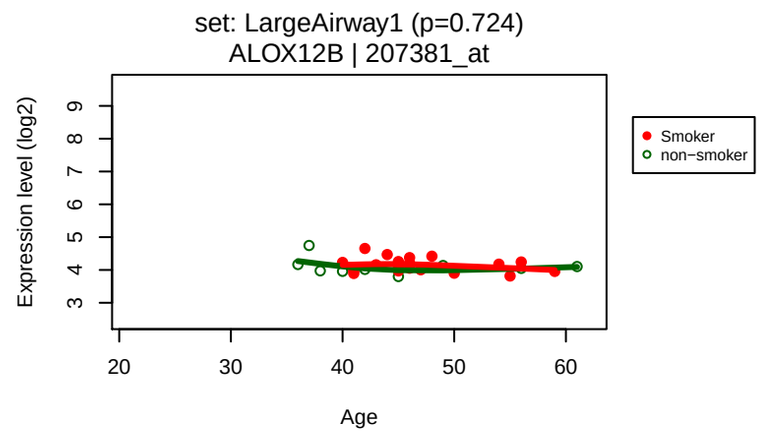

Gene expression of smokers and nonsmokers
gene: ALOX12B | 207381_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0433
p=0.0659
p=0.0306
p=0.724
p=0.6231
p=0.0075
p=0.7932
p=0.3573
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

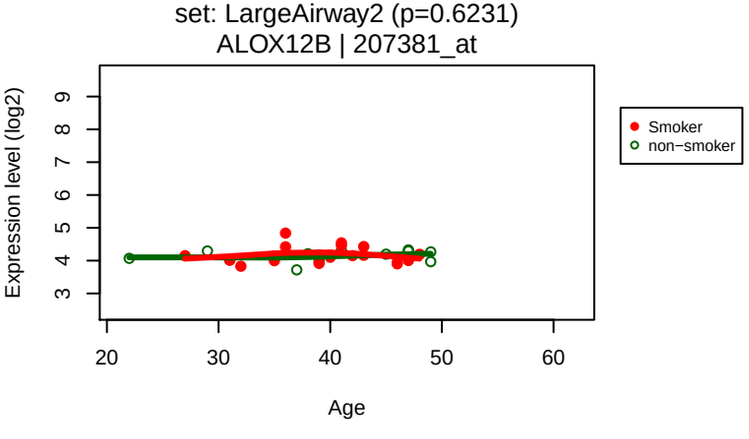

Gene expression ratio (smokers/nonsmokers)
gene: ALOX12B | 207381_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0433
p=0.0659
p=0.0306
p=0.724
p=0.6231
p=0.0075
p=0.7932
p=0.3573
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

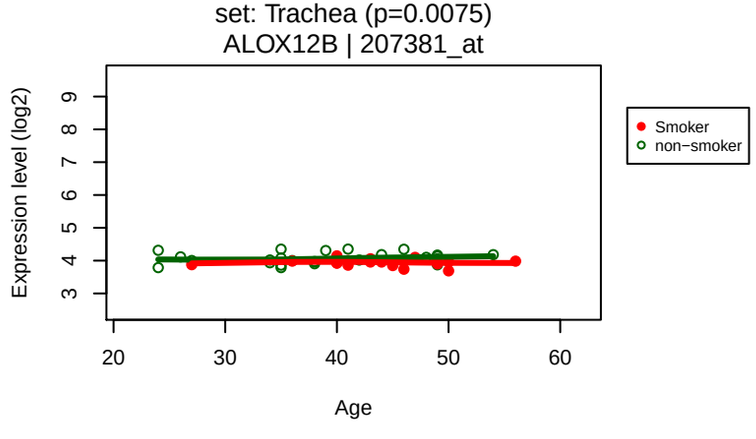

Gene expression ratio in Males
gene: ALOX12B | 207381_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0651
p=0.1417
p=0.0439
p=0.5632
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALOX12B | 207381_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7129
p=0.039
p=0.1201
p=0.447
SmallAirway2
LargeAirway2
Trachea
Macrophage2

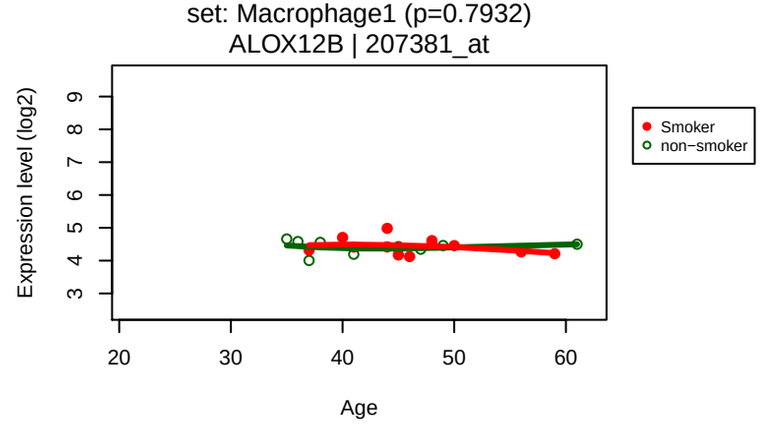

set: SmallAirway1 (p=0.0433)
ALOX12B | 207381_at
Expression level (log2)
Age
Smoker
non−smoker

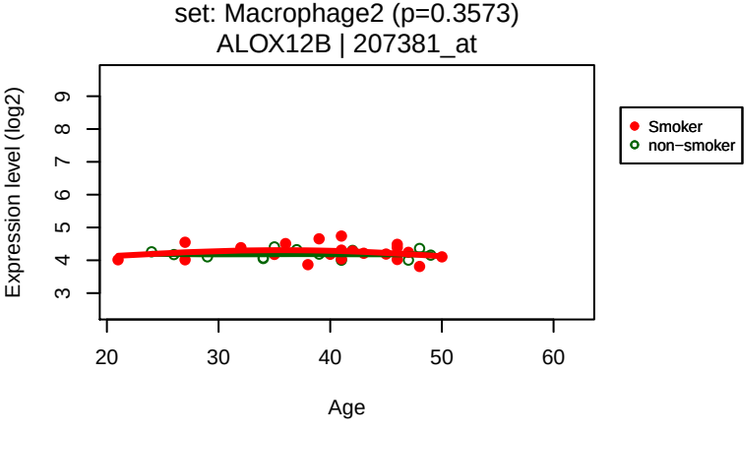

set: SmallAirway2 (p=0.0659)
ALOX12B | 207381_at
Expression level (log2)
Age
Smoker
non−smoker

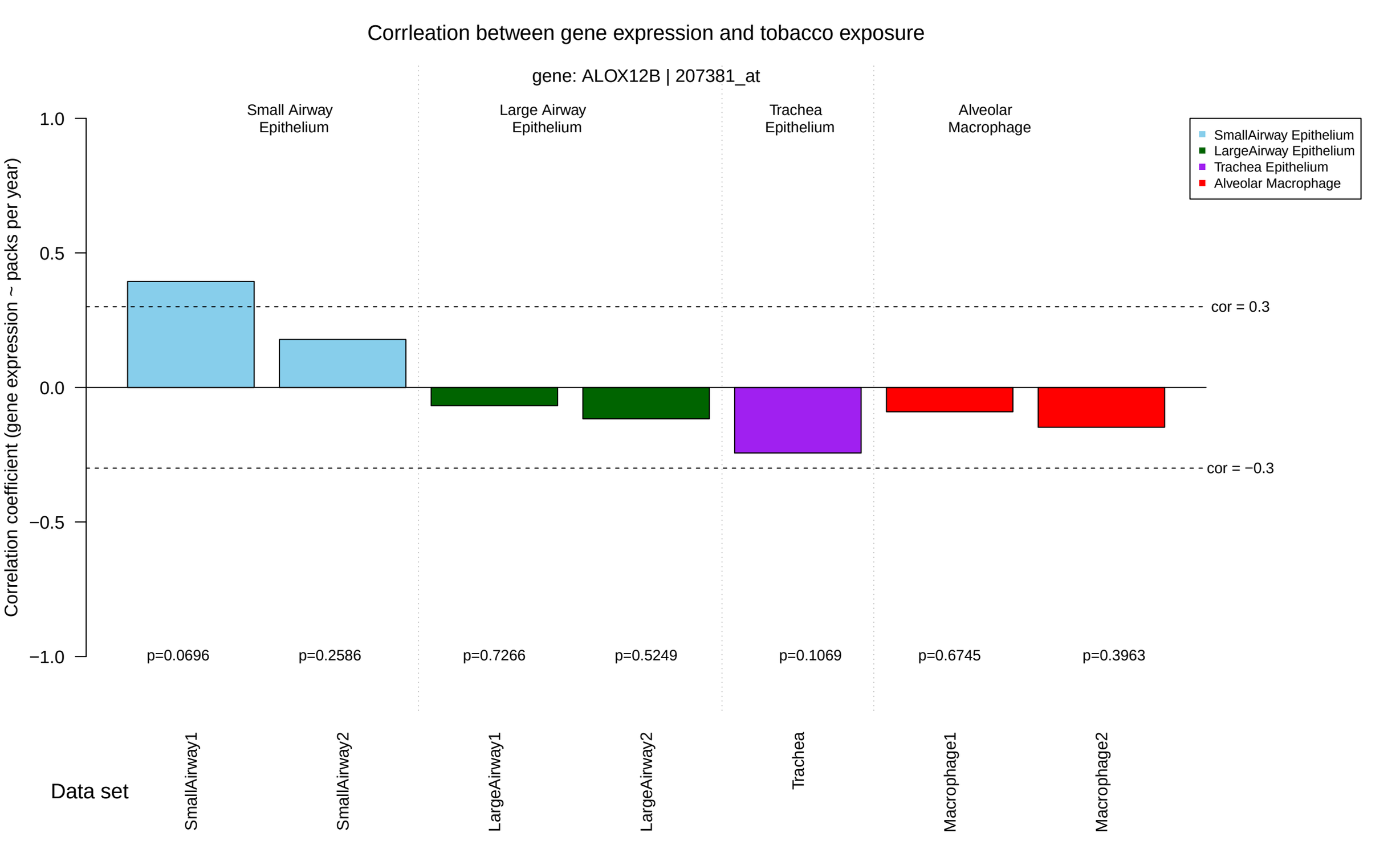

set: LargeAirway1 (p=0.724)
ALOX12B | 207381_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6231)
ALOX12B | 207381_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0075)
ALOX12B | 207381_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7932)
ALOX12B | 207381_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3573)
ALOX12B | 207381_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALOX12B | 207381_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0696
p=0.2586
p=0.7266
p=0.5249
p=0.1069
p=0.6745
p=0.3963
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2